



Rajiv Gandhi Institute of Petroleum Technology, Jais Amethi

Department of Computer Science and Engineering

Object-Oriented Methodologies Lab (CS224L)

Assignment 5

Q.1 Create a class with a data member with a “serial number” for each object created from the class. That is, the first object created will be numbered 1, the second 2 and so on. To do this, you will need another data member that records a count of how many object have been created so far. (This member should apply to the class as a whole; not to individual objects. What keyword specifies this?) Then, as each object is created, its constructor can examine this count member variable to determine the appropriate serial number for the new object. Add a member function that permits an object to report its own serial number. Then write a main () program that creates three objects and quires each one about its serial number. They should respond I am object number 2, and so on.

Q.2 Create a class time that has separate int member data for hours, minutes and seconds. One constructor should initialize this data to 0, and another should initialize it to fixed values. Another member function should display it, in 11:59:59 format. The final member function should add two objects of type time passed as arguments. A main program should create two initialized time objects (should they be const?) and one that is not initialized. Then it should add the two initialized values together, leaving the result in the third time variable. Finally, it should display the value of this third variable. Make appropriate member functions const.
Q.3 Write a program to find the grace marks for a student using switch. The user should enter the class obtained by the student and the number of subjects he has failed in. Use the following logic:

Q.3 In ocean navigation, locations are measured in degrees and minutes of latitude and longitude. Thus if you’re lying off the mouth of Papeete Harbor in Tahiti, your location is 149 degrees 34.8 minutes west longitude, and 17 degrees 31.5 minutes south latitude. This is written as 149°34.8’ W, 17°31.5’ S. There are 60 minutes in a degree. (An older system also divided a minute into 60 seconds, but the modern approach is to use decimal minutes instead.) Longitude is measured from 0 to 180 degrees, east or west from Greenwich, England, to the international dateline in the Pacific. Latitude is measured from 0 to 90 degrees, north or south from the equator to the poles. Create a class angle that includes three member variables: an int for degrees, a float for minutes, and a char for the direction letter (N, S, E, or W). This class can hold either a latitude variable or a longitude variable. Write one-member function to obtain an angle value (in degrees and minutes) and a direction from the user, and a second to display the angle value in 179°59.9’ E format. Also write a three-argument constructor. Write a main() program that displays an angle initialized with the constructor, and then, within a loop, allows the user to input any angle value, and then displays the value.